

Practice Set 7

MA628: Financial Derivatives Pricing

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Q.1 Let $X_1, X_2, \dots$ be iid RVs integrable and such that $\mathbb{E}(X_1) = 0$ and $\text{var}(X_1) = \sigma^2$. Let $\mathcal{F}_n = \sigma(X_1, X_2, \dots, X_n)$ be the σ -algebra generated by random variables $X_1, X_2, \dots, X_n$. Further, let

$$S_n = \sum_{j=1}^n X_j.$$

Prove that following are martingale.

- (a) S_n
- (b) $M_n = S_n^2 - n\sigma^2$
- (c) Suppose $\mathbb{E}X_1 = 1$, show that $M_n = \prod_{j=1}^n X_j$ is a martingale.
- (d) $M_n = \mathbb{E}(X|\mathcal{F}_n)$ for some random variable X .
- (e) $M_n = f(\lambda)^{-n}e^{\lambda S_n}$, where $f(\lambda) = \mathbb{E}(e^{\lambda X_1})$, $\lambda \neq 0$.

Q.2 Let N_t , $t \geq 0$ is a Poisson process with intensity $\lambda > 0$. Show that $N_t - \lambda t$ is a martingale.

Q.3 Let $X_t = \sum_{i=1}^{N_t} Y_i$ and N_t be a Poisson process with intensity $\lambda > 0$. Suppose Y_i are i.i.d. random variables independent of N_t , having $N(\mu, \sigma^2)$. Find $m > 0$ such that $X_t - mt$ is a martingale.

Q.4 Show that the following functions of Brownian motion are martingale with respect to the natural filtration.

- (a) $B(t)^3 - 3tB(t)$
- (b) $B(t)^4 - 6tB(t)^2 + 3t^2$.

Q.5 Show that the solution of Bachelier model given by $dS(t) = \sigma dB(t)$ is a martingale, where $S(t)$ represents the equity price. Further, suppose 10 consecutive days equity prices for some equity are given (in Rs) as

100, 105, 102, 110, 117, 104, 107, 101, 120, 115.

Estimate σ , using this data.

Q.6 Consider the modified *Bachelier Model* for equity price $dS(t) = \mu dt + \sigma dB(t)$, where $S(t)$ and $B(t)$ represents the equity price and standard Brownian motion respectively. Write the solution of this differential equation and the procedure to estimate the parameters μ and σ . Suppose for an equity 10 days consecutive prices are 100, 105, 102, 107, 109, 108, 106, 114, 115, 116. Calculate the estimates of μ and σ using this data.

Q.7 For the exponential model of equity price

$$S(t) = e^{a+bt+cB(t)},$$

where $B(t)$ is standard Brownian motion, estimate the parameters a , b and c using the equity prices data for 10 consecutive days: 100, 105, 102, 107, 109, 108, 106, 114, 115, 116.

Q.8 Find the probability density function (PDF) of $B(1) + B(2) + B(3) + \cdots + B(n)$, $n \geq 2$, where $B(t)$ is standard Brownian motion.